Smart Lead Management

&

Routing System

Technical Documentation

Developed by

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Automation Platform

- Make.com

Supporting Services

- Gmail
- Google Sheets

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1. Introduction

The Smart Lead Management & Routing System is a no-code automation workflow developed using Make.com.

The system automates lead processing from initial submission through validation, enrichment, scoring, routing, notification, scheduling, storage, and logging.

The workflow integrates Google Forms, Google Sheets, Make.com, Google's DNS API, Gmail, and Google Calendar.

2. System Objective

The objective is to automate the processing of incoming leads while ensuring that each valid lead is:

1. Captured
2. Validated
3. Enriched
4. Processed
5. Scored
6. Routed
7. Stored
8. Communicated to staff
9. Scheduled for follow-up
10. Logged

3. High-Level Architecture

Google Form

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Google Sheets

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Make — Watch New Rows

↓

Lead Validation

↓

Initialize Lead Status

↓

HTTP DNS/MX Enrichment

↓

Lead Processing

↓

Lead Score Components

↓

Calculate Lead Score

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Router

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Sales Nurture

↓ ↓

Storage

↓ ↓

Gmail

↓ ↓

Calendar

↓ ↓

Activity Log

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4. Data Flow

Stage 1 — Lead Submission

The lead submits the Google Form.

The following fields are collected:

- Timestamp
- Full Name
- Email Address
- Company Name
- Job Title
- Company Size
- Area of Interest
- Estimated Budget

The submission is written to Google Sheets.

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5. Make Trigger

Module

Google Sheets — Watch New Rows

The trigger monitors the Google Form response sheet.

When a new row appears, Make.com creates a bundle containing the lead information.

Reason for using Google Sheets

The original architecture considered a Custom Webhook.

However, Google Forms does not directly POST form submissions to an arbitrary Make webhook.

Therefore:

Google Form

↓

Google Sheets

↓

Watch New Rows

was selected.

The assessment permits an app-based trigger.

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6. Lead Validation

Module

Filter — Lead Validation

The filter checks that the following fields exist:

Full Name

Email Address

Company Name

Job Title

Company Size

Area of Interest

Estimated Budget

The conditions are connected using:

AND

Therefore, a lead must contain all required fields before continuing.

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7. Initialize Lead Status

Module

Tools — Set Variables

Variables:

lead_status

email_domain

lead_status

Initial value:

New

email_domain

The email domain is extracted from the submitted email address.

Example:

someone@example.com

↓

example.com

This value is later passed to the DNS API.

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8. External API Enrichment

Module

HTTP — Make a Request

API

Google DNS-over-HTTPS

https://dns.google/resolve

Method

GET

Query Parameters

name = email_domain

type = MX

Parse Response

Yes

The API returns DNS information for the requested domain.

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9. MX Verification

The system examines whether MX records are returned for the email domain.

A successful response indicates that mail-server records were found.

The workflow stores this information as:

mx_verified = Yes

This is an email-domain verification/enrichment step.

It should not be interpreted as complete company verification.

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10. Lead Processing

Module

Tools — Set Variables

Variables:

lead_priority

processed_at

mx_verified

Example:

lead_priority = Pending

processed_at = current timestamp

mx_verified = Yes

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11. Lead Score Components

The workflow calculates three separate scoring components.

Budget

$5,000–$9,999 → 30

$1,000–$4,999 → 20

Other → 10

Company Size

100–249 → 25

50–99 → 20

Other → 15

Area of Interest

AI / Machine Learning → 30

Data Analytics → 25

Software Development → 20

Other → 10

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12. Total Lead Score

The final score is calculated as:

Lead Score =

Budget Score

+

Company Size Score

+

Interest Score

Example:

Budget Score = 30

Company Size Score = 25

Interest Score = 30

Lead Score = 85

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13. Router Logic

The router has two branches.

Sales Branch

Condition:

Lead Score >= 60

Nurture Branch

Condition:

Lead Score < 60

This ensures that each lead follows only the branch corresponding to its score.

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14. Sales Branch

The Sales branch contains:

1. Google Sheets — Add a Row

Stores the processed lead in the Lead Results sheet.

2. Gmail — Send an Email

Sends a Sales notification containing the lead information and score.

3. Google Calendar — Create an Event

Creates a Sales follow-up event.

4. Google Sheets — Add a Row

Records the action in the Activity Log.

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15. Nurture Branch

The Nurture branch contains:

1. Google Sheets — Add a Row

Stores the processed lead.

2. Gmail — Send an Email

Sends a Nurture notification.

3. Google Calendar — Create an Event

Creates the Nurture follow-up.

4. Google Sheets — Add a Row

Records the action in the Activity Log.

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16. Lead Results Data Structure

The Lead Results sheet contains:

Column	Purpose
Timestamp	Submission time
Full Name	Lead name
Email	Lead email
Company	Company name
Job Title	Lead's job title
Company Size	Organization size
Area of Interest	Lead interest

Column	Purpose
Budget	Estimated budget
Email Domain	Extracted domain
MX Verified	Enrichment result
Lead Score	Calculated score
Route	Sales/Nurture
Status	Processing status

17. Activity Log Data Structure

The Activity Log contains:

Column	Purpose
Timestamp	Time of action
Lead Email	Lead identifier
Action	Action performed
Route	Sales/Nurture
Score	Lead score
Status	Lead status
Details	Additional information

Example:

Action:

Sales Lead Routed

Route:

Sales

Details:

Sales alert sent and follow-up scheduled

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18. Error Handling

The HTTP enrichment module contains a Resume error handler.

If the external DNS request fails, the handler provides fallback data instead of immediately terminating the scenario.

This demonstrates the use of failure handling in an external API integration.

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19. Testing Strategy

The workflow was tested using both routing conditions.

Test Case 1 — Nurture

Expected:

Lead Score < 60

Observed:

Lead Score = 35

Route = Nurture

The Nurture branch executed successfully.

Test Case 2 — Sales

Expected:

Lead Score >= 60

Observed:

Lead Score = 85

Route = Sales

The Sales branch executed successfully.

The downstream actions included:

- Lead Results storage
- Gmail notification
- Google Calendar event
- Activity Log entry

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20. Authentication & Permissions

The workflow requires authenticated connections to:

- Google Sheets
- Gmail
- Google Calendar

During development, Google Calendar initially returned a permissions error.

The connection permissions were updated, allowing the Calendar module to create events successfully.

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21. Challenges Encountered

Challenge 1 — Trigger Selection

The initial webhook approach was reconsidered because Google Forms does not directly POST to a Make webhook.

Resolution

Google Sheets Watch New Rows was selected.

Challenge 2 — Email Domain Extraction

The initial expression for extracting the domain caused an error.

Resolution

The expression was corrected to extract the final section after the @ symbol.

Challenge 3 — HTTP 400 Error

The DNS API initially returned an HTTP 400 error because the query parameter was not being passed correctly.

Resolution

The email-domain variable was corrected before being passed into the HTTP request.

The API subsequently returned a successful response.

Challenge 4 — Scoring Formula

The initial scoring logic returned default values instead of the intended scores.

Resolution

The comparison expressions were corrected.

The resulting scoring logic produced the expected values.

Challenge 5 — Sales Routing

The Sales branch initially did not execute because the calculated score was below the threshold.

After correcting the scoring formulas, a score of 85 successfully entered the Sales route.

Challenge 6 — Calendar Authorization

Google Calendar initially returned an authorization error.

Resolution

The Google connection was re-authorized with the required permissions.

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22. Design Decisions

Why Google Sheets?

Google Sheets provides a simple interface for collecting, viewing, and storing lead data while integrating directly with Make.com.

Why Rule-Based Scoring?

A deterministic scoring model was selected because it is transparent, easy to test, and appropriate for this assessment.

Why a Router?

The router allows a single automation to handle multiple lead outcomes.

Why an Activity Log?

The Activity Log provides traceability and makes it possible to review the actions performed by the automation.

Why Error Handling?

External API requests can fail, so the enrichment step requires a recovery path.

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23. Limitations

The current system has several limitations.

Email enrichment

The DNS lookup checks MX records but does not verify the identity of the individual or company.

Rule-based scoring

The score is based on predefined rules rather than historical conversion data.

No CRM

The current workflow stores results in Google Sheets rather than a dedicated CRM.

No duplicate detection

Duplicate lead submissions are not currently filtered.

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24. Future Improvements

Future versions could introduce:

CRM Integration

Automatically create contacts, leads, or deals in a CRM.

Company Enrichment

Retrieve company information such as:

- Industry
- Employee count
- Revenue

- Website
- Location

Duplicate Detection

Check whether a lead already exists before creating a new record.

Advanced Lead Scoring

Use additional factors and eventually historical conversion data.

AI-Assisted Classification

Use an AI model to classify lead intent or summarize lead information.

Team Collaboration

Integrate Slack or Microsoft Teams for real-time notifications.

Analytics

Create a dashboard showing:

- Lead volume
- Sales-qualified leads
- Nurture leads
- Average score
- Lead sources
- Follow-up status

Monitoring

Create centralized automation monitoring and error reporting.

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25. Security Considerations

Private credentials, API keys, authentication tokens, and private webhook URLs should never be committed to GitHub.

The repository should contain only safe documentation and demonstration materials.

Test data should be used where possible.

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26. Project Outcome

The completed automation demonstrates an end-to-end lead workflow using Make.com.

The workflow successfully demonstrates:

- Automated data capture
- Validation
- Data transformation
- API integration
- Rule-based scoring
- Conditional routing
- Email notifications
- Calendar scheduling
- Data storage
- Activity logging
- Error handling

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27. Repository Documentation

Additional project materials are organized as follows:

/smart-lead-management-routing/

```
│
├── README.md
│
├── docs/
│   └── technical-documentation.pdf
│
├── screenshots/
│   ├── Google Form.png
│   ├── Form Responses sheet.png
│   ├── Lead Results sheet.png
│   ├── Activity Log sheet.png
│   └── Complete Make.com scenario.png
│
├── demo/
│   └── drive.google.com
│
└── LICENSE
```

28. Demo

A four-minute demonstration accompanies this project.

The demonstration covers:

1. Lead capture
2. Validation
3. Enrichment
4. Scoring
5. Routing
6. Notification
7. Scheduling
8. Logging

Demo:

[https://drive.google.com/file/d/1IDcClrsq6TlyjF7CBh0Cv7lHw4Pwmjfc/view?usp=sharing]

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29. Author

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My professional interests combine healthcare, laboratory science, data, computational biology, and technology.

Areas of interest include:

- Medical Laboratory Science
- Data Analytics
- Data Science
- Bioinformatics
- Computational Biology
- Computational Drug Discovery
- Healthcare Technology
- Automation

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31. Conclusion

The Smart Lead Management & Routing System demonstrates how no-code automation can connect multiple business processes into a single workflow.

Rather than manually processing every lead, the system automatically validates, enriches, scores, routes, communicates, schedules, stores, and logs lead activity.

The project also demonstrates practical problem-solving during automation development, including API troubleshooting, formula debugging, authentication issues, routing validation, and workflow testing.